

SWATHI PADMANABHAN

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Website

Experience

Department of Bioengineering, Imperial College London

ICRF-Schmidt AI in Science Postdoctoral Fellow

August, 2026 – Present

London, UK

TANUH (Translational AI for Networked Universal Healthcare) AICoE

Research Staff Member

January, 2026 – July, 2026

Bangalore, India

Education

Indian Institute of Science

PhD and M.Tech. (by Research), Department of Instrumentation and Applied Physics

Aug. 2019 – Sept, 2025

Bangalore, India

L.B.S Institute of Technology for Women

Bachelors of Technology in Applied Electronics and Instrumentation

Aug. 2015 – June 2019

Thiruvananthapuram, India

Selected Fellowships and Recognition

1. **Imperial College Research Fellowship - AI in Science (supported by Schmidt Sciences)**, Imperial College London (2025) [[Offer Letter](#)]
2. **ETH Zurich Postdoctoral Fellowship**, ETH Zurich, 2025. [Declined] [[Offer Letter](#)]
3. **Seal of Excellence for Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship**, European Commission Horizon Europe, 2025. [[Seal Of Excellence](#)]

Publications

1. **Swathi Padmanabhan** and Jaya Prakash, “Deep Tissue Sensing of Chiral Molecules using Polarization Enhanced Photoacoustics,” Science Advances, 11, eado8012 (2025). DOI:10.1126/sciadv.ado8012. [[Journal Article](#)] [[News Coverage](#)]
2. **Swathi Padmanabhan**, Sarikonda Aryan Shashank, Rajesh Srinivasan and Jaya Prakash, “Optoacousto-fluidics based optical rotation measure for blood glucose sensing,” vol. 72, no. 11, pp. 3312-3323, Nov. 2025, doi: 10.1109/TBME.2025.3566940. [[Journal Article](#)]
3. **Swathi Padmanabhan** and Jaya Prakash, “Detecting optical rotary dispersion from photoacoustic circular dichroism of glucose in the NIR-II wavelength”, IEEE Journal of Selected Topics in Quantum Electronics, doi: 10.1109/JSTQE.2025.3610710 [[Journal Article](#)]
4. **Swathi Padmanabhan** and Jaya Prakash, “Optimal Pathlength Identification for Accurate Glucose Sensing with Photoacoustic Derived Optical Rotation,” Optics Letters, 50, 149-152 (2025). [[Journal Article](#)]
5. Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash and V.R. Supradeepa, “Pulsed cascaded Raman fiber laser widely tunable in the second near-infrared and visible window for hyperspectral photoacoustic imaging”, Optics Letters, 50, 2223-2226 (2025). [[Journal Article](#)][[Optica Spotlight](#)]
6. **Swathi Padmanabhan***, Madhavi Tripathi*, Jaya Prakash and Ashok Raichur, “Seed mediated galvanic synthesis of ultrasmall CuS-Au nanohybrids for photo-theranostic applications,” ACS Applied Nano Materials 6, 14861-14875 (2023). [[Journal Article](#)] [[News Coverage](#)]
7. Atheeth. S and Chandrashekar L N and Isha Munjal and **Swathi Padmanabhan** and Jaya Prakash and Manish Arora, “Microfabrication of a 16 MHz 1D-pMUT-Array for photoacoustic and ultrasound imaging,” Ultrasonic Imaging. 2025;0(0). doi:10.1177/01617346251380297 [[Journal Article](#)]

8. Hendry Saju, Sharon George, **Swathi Padmanabhan**, Nisarga Ramesh, Tharun Jaikumar, Victor Mukherjee, Jaiprakash N. Sangshetti, Shajesh Palantavida, Nandakumar Dalavaikodihalli Nanjaiah, Jaya Prakash, Chulhong Kim, Rohidas B. Arote; "*Harnessing Rare-Earth Materials as Exogenous Contrast Agents for Enhanced Photoacoustic Imaging.*" Chemical Biomedical Imaging 2026; <https://doi.org/10.1021/cbmi.5c00280> [[Journal Article](#)]
9. Ovindu Chakrawarthige, Naveen Paluru, **Swathi Padmanabhan**, Aaqilah AJ, Gowry Nandana M, Utkarsh Gupta, Raji Susan Mathew and Navchetan Awasthi, "*Knowledge Distillation in Medical Image Analysis: A Survey*", Array [[In Press](#)]

* denotes equal contribution

Patent

1. **Swathi Padmanabhan** and Jaya Prakash, "*Apparatus for measuring concentrations of chiral molecules*" Indian Patent; Patent Number: IN547866 (2024). [[Granted](#)] [[Patent](#)]

Book Chapters

1. Anandhu Anil Sindu, **Swathi Padmanabhan**, and Jaya Prakash. "Emerging Approaches in Metabolic Imaging Using Photoacoustics." Bioimaging Modalities in Bioengineering, vol 9. Springer, Cham (2025): 529-545. [[Book Chapter](#)]
2. **Swathi Padmanabhan** and Uma Ranjan, "*Markov Chain Monte Carlo*," In: Daya Sagar, B.S., Cheng, Q., McKinley, J., Agterberg, F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham (2022). [[Book Chapter](#)]

Conferences and Workshops

- Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash, V. R. Supradeepa, "Photoacoustic Spectroscopy Using a Widely Tunable Pulsed Cascaded Raman Fiber Laser," 2025 Conference on Lasers and Electro-Optics Europe & European Quantum Electronics Conference (CLEO/Europe-EQEC), Munich, Germany, 2025, pp. 1-1. (23-27 June 2025) [[Conference Abstract](#)]
- Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash, V. R. Supradeepa, "Passively Q-switched Raman fiber laser source widely tunable in near infrared and visible window," Proc. SPIE 13347, Nonlinear Frequency Generation and Conversion: Materials and Devices XXIV, 133470T (21 March 2025). [[Conference Paper](#)]
- Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash, V.R. Supradeepa, "Widely tunable Raman fiber laser for hyperspectral photoacoustic imaging," Proc. SPIE 13319, Photons Plus Ultrasound: Imaging and Sensing 2025, 1331911 (20 March 2025). [[Conference Paper](#)]
- **Swathi Padmanabhan** and Jaya Prakash, "*Approaching Optical Rotation Through the Lens of Near-Infra-Red-II Photoacoustics*," Optical Tomography and Spectroscopy, OTu3D. 6, Optica Biophotonics Congress: Biomedical Optics (2024). [[Conference Abstract](#)] [[Oral Presentation](#)]
- **Swathi Padmanabhan** and Jaya Prakash, "*Exploring Photoacoustics from Optical Activity: A Perspective for Biomolecular Sensing*," 2024 IEEE South Asian Ultrasonics Symposium (SAUS) (2024). [[Oral Presentation](#)]
- **Swathi Padmanabhan** and Jaya Prakash, "*Mathematical Modelling for Polarization Based Photoacoustic System to Evaluate the Changes in Propagation of Polarized Light in Tissues*," International Conference on Photonics, Photonics (2023). [[Conference Poster](#)]
- **Swathi Padmanabhan** and Jaya Prakash, "*Mathematical Modelling for Polarization Based Photoacoustic System to Evaluate the Changes in Propagation of Polarized Light in Tissues*," Workshop on Medical Image

Computing, Indian Institute of Science (2023). [[Workshop Poster](#)]

- **Swathi Padmanabhan** and Jaya Prakash, “*Development of Polarisation based Photoacoustic Imaging System and Appropriate Mathematical models to enable the Propagation of Polarized Light in Tissues*”, Workshop on The role of AI in Transforming Healthcare, Indian Institute of Science (2022). [[Workshop Poster](#)]

Relevant Coursework

- Biomedical Optics and Spectroscopy
- Medical Imaging
- Optical Metrology
- Digital Image Processing
- Advanced Image Processing
- Numerical Methods
- Sensors and Transducers
- Biology for Engineers
- Drug Delivery

Technical & Experimental Expertise

- Microfluidic Channel Design & Fabrication (Soft Lithography)
- CAD designing for 3D printing
- Instrumentation - optomechanics, system calibration, motion actuator control
- Ultrasound system response characterisation
- Optics - Laser Safety, Optical alignment, Polarization optics, Laser energy calibration
- Operation of Pulsed lasers, diode lasers, CW lasers and Raman Fiber lasers
- Electronics -Data acquisition with Arduino, Raspberry Pi & TI AFE5818EVM evaluation boards.
- Cell Culture (Beginner’s level) - HeLa Cells
- Operation of UV-VIS Spectrophotometer
- Operation of Oxygen Plasma for bonding techniques
- in fabrication
- Operation of US single element, linear array and PMUT transducers for US data acquisition
- Devising Human Ethics protocols
- Patient studies and involvement with doctors and clinicians
- Signal conditioning for post-processing of data
- Mathematical formulations for acoustic signal analysis for optical concepts
- Ultrasound and optical Phantom preparation
- Post-processing of experimental Ultrasound data
- Handling Patient data and samples
- *Ex-vivo* experimental design for therapy and sensing
- *In-vivo* experimental optimisation
- Implementation of Polarised Monte Carlo with depthwise optical activity

Technical Skills

Languages: C, MATLAB, Python, C++

Developer Tools: Visual Studio Code, LabVIEW

Academic Service/Outreach

Reviewer: IEEE Transactions on Medical Imaging, IEEE Transactions on Circuits and Systems I, Journal of Biomedical Optics, IEEE Photonics, Physica Scripta

Teaching: Teaching Assistance for courses Biomedical Optics and Spectroscopy, Medical Imaging

Outreach: Institute (IISc) Open Day demos, Outreach with high school students on prospective higher education

Professional Societies: Student Member of SPIE and Optica. Student Member of Applied Photonics Initiative (API) at IISc.